

Lecture #3, Sep 16, 2025

Finite difference approximation of PDEs

(See Ch1 LeVeque)

- 1) How to derive formula and error term analytically
- 2) How to simulate approximations and errors

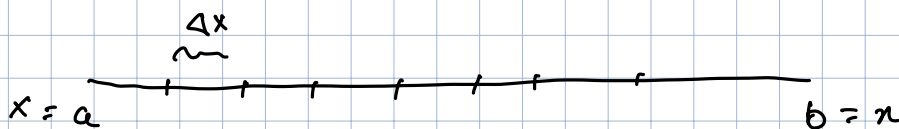
Finite difference approximation of $u(x, t)$

where $x \in [a, b]$, $t \in [0, T]$, $T > 0$

let $x_j = j \Delta x$ for $j = 0, \dots, N$

$t_n = n \Delta t$ for $n = 0, \dots, M$

$$\begin{array}{l} \Delta x \text{ is space step} \\ \Delta t \text{ is time step} \end{array} \quad \left| \begin{array}{ll} x_0 = a, & x_N = b \\ t_0 = 0, & t_M = T \end{array} \right| \quad \begin{array}{l} \Delta x = x_{j+1} - x_j \\ \Delta t = t_{n+1} - t_n \end{array}$$

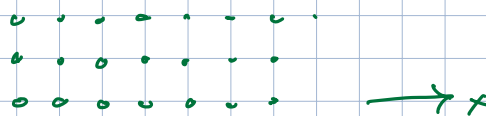


Want $u(x, t)$, $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial t}$ etc. at (x_j, t_n)

ignore t for now.

$\uparrow t$

Collection of (x_j, t_n) is
called a grid or mesh



"Given $u(x, t)$ at one point(s)

want to estimate $u(x, t)$ at other points."

Approximate $\frac{\partial u}{\partial x}$:

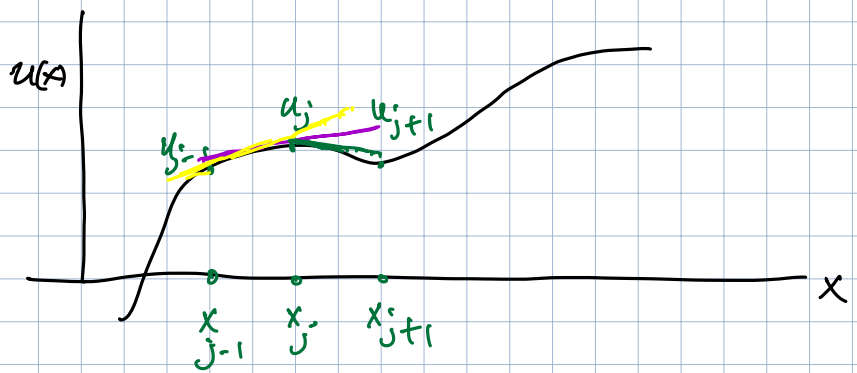
Note $x_{j+1} = x_j + \Delta x$

$x_{j-1} = x_j - \Delta x$

$u(x_j) = u_j$

$u(x_{j+1}) = u_{j+1}$

$u(x_{j-1}) = u_{j-1}$



$$\frac{\partial u}{\partial x} \text{ at } x_j = \frac{\text{rise}}{\text{run}} \approx \frac{u_{j+1} - u_j}{x_{j+1} - x_j} \quad \text{or} \quad \frac{u_j - u_{j-1}}{x_j - x_{j-1}} \quad \text{or} \quad \frac{u_{j+1} - u_{j-1}}{x_{j+1} - x_{j-1}}$$

FD formula

$$\left. \frac{\partial u}{\partial x} \right|_j = \begin{cases} \frac{u_{j+1} - u_j}{\Delta x} & \text{forward} \\ \frac{u_j - u_{j-1}}{\Delta x} & \text{backward} \\ \frac{u_{j+1} - u_{j-1}}{2\Delta x} & \text{Central} \end{cases}$$

FD operator

$$\delta_+ u = u_{j+1} - u_j, \quad D_+ u = \frac{\delta_+ u}{\Delta x}$$

$$\delta_- u = u_j - u_{j-1}, \quad D_- u = \frac{\delta_- u}{\Delta x}$$

$$\delta_c u = u_{j+1} - u_{j-1}, \quad D_c u = \frac{\delta_c u}{2\Delta x}$$

Error def.

let $\{u_0, u_1, \dots, u_N\}$ be exact values

$\{v_0, v_1, \dots, v_N\}$ be approximations

error $e_j = |u_j - v_j|$ for $j = 0, 1, \dots, N$

measure or norm

Infinity or max norm $\|e\|_{\infty} = \max_j |e_j|$

1-norm $\|e\|_1 = \sum_{j=0}^N |e_j|$

2-norm $\|e\|_2 = \sqrt{\sum_{j=0}^N |e_j|^2}$ (r.m.s error)

We can see that

$$\|e\|_{\infty} \leq \|e\|_2 \leq \sqrt{n} \|e\|_{\infty}$$

[for $\|e\|_m, \|e\|_n, \quad c_1 \|e\|_m \leq \|e\|_n \leq c_2 \|e\|_m$]

Ex: Calculate FD approximation of $\frac{\partial u}{\partial x}$

where $u(x) = \sin 2\pi x, \quad x \in [0, 1]$

Truncation error : Taylor series

$$u_{j+1} = u(x_j + \Delta x) = u(x_j) + \Delta x \frac{\partial u}{\partial x} \Big|_j + \frac{\Delta x^2}{2!} \frac{\partial^2 u}{\partial x^2} \Big|_j + \dots$$

$$\underbrace{\frac{u_{j+1} - u_j}{\Delta x} = \frac{\partial u}{\partial x} \Big|_j}_{\text{FD formula}} + \underbrace{\frac{\Delta x}{2!} \frac{\partial^2 u}{\partial x^2} \Big|_j + \frac{\Delta x^2}{3!} \frac{\partial^3 u}{\partial x^3} \Big|_j + \dots}_{\text{error}}$$

This is how we derive FD formula.

The lowest order in Δx term in error is called leading order error.

$$\text{lead error} = \frac{\Delta x}{2!} \frac{\partial^2 u}{\partial x^2} \Big|_j = O(\Delta x) \text{ or 1st order.}$$

$$\frac{\partial u}{\partial x} \Big|_j = \frac{u_{j+1} - u_j}{\Delta x} - \underbrace{\frac{\Delta x}{2!} \frac{\partial^2 u}{\partial x^2} \Big|_j}_{\text{lead error}} - \dots$$



We say that FD approximation is consistent if all error terms goes to zero if $\Delta x \rightarrow 0$

Notice $\lim_{h \rightarrow 0} \frac{f(h+a) - f(a)}{h} = \frac{\partial f}{\partial x}(a)$

Remark There exist constants C and p such that the error follows

$$E(\Delta x) \approx C \Delta x^p \quad \text{for a general FD formula.}$$

$$\text{or } \log |E(\Delta x)| = p \log \Delta x + \log C.$$

We say the method is 1st order if $p=1$

2nd " " if $p=2$

and so on.

Ex: Derive central FD formula and show that it is second order accurate

Sol: Taylor series $\Rightarrow$

$$u_{j+1} = u_j + \Delta x \frac{\partial u}{\partial x} \Big|_j + \frac{\Delta x^2}{2!} \frac{\partial^2 u}{\partial x^2} + \dots \quad (1)$$

$$u_{j-1} = u_j - \Delta x \frac{\partial u}{\partial x} \Big|_j + \frac{\Delta x^2}{2!} \frac{\partial^2 u}{\partial x^2} - \dots \quad (2)$$

① - ② $\Rightarrow$

$$u_{j+1} - u_{j-1} = + 2\Delta x \frac{\partial u}{\partial x} \Big|_j + 2 \frac{\Delta x^3}{3!} \frac{\partial^3 u}{\partial x^3} \Big|_j + \dots$$

$$\frac{u_{j+1} - u_{j-1}}{2\Delta x} = \frac{\partial u}{\partial x} \Big|_j + \frac{\Delta x^2}{3!} \frac{\partial^3 u}{\partial x^3} \Big|_j + \dots$$

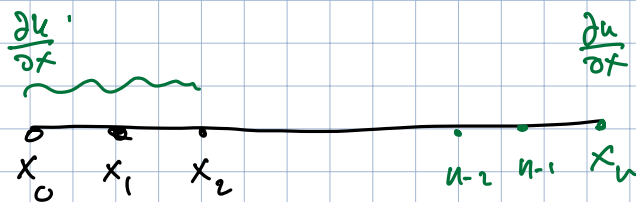
$$\text{lead error} = \frac{\Delta x^2}{3!} \frac{\partial^3 u}{\partial x^3} \Big|_j$$

$$= O(\Delta x^2)$$

Ex: Derive FD formula ^{for $\frac{\partial u}{\partial x}$} using $\{x_{j-1}, x_j, x_{j+1}\}$
and find lead error.

Repeat: using $\{x_j, x_{j+1}, x_{j+2}\}$

Repeat using $\{x_{j-2}, x_{j-1}, x_j\}$



Lecture #4, Sep 18, 2025

Continue on FD approximations

- Derivation of higher order approximations
- Dispersion relation of schemes

Generalization (method of undetermined coefficients)

Ex. Derive FD formula for $\frac{\partial u}{\partial x}$ at x_j using $\{x_{j-1}, x_j, x_{j+1}\}$ (symmetric stencil)

Solⁿ

$$\frac{\partial u}{\partial x}|_j = a u_{j-1} + b u_j + c u_{j+1}$$

$$= a \left[u_j - \Delta x \frac{\partial u}{\partial x}|_j + \frac{\Delta x^2}{2!} \frac{\partial^2 u}{\partial x^2}|_j - \dots \right]$$

$$+ b u_j$$

$$+ c \left[u_j + \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2!} \frac{\partial^2 u}{\partial x^2} + \dots \right]$$

$$\frac{\partial u}{\partial x}|_j = (a+b+c) u_j + (-a+c) \Delta x \frac{\partial u}{\partial x} + (a+c) \frac{\Delta x^2}{2!} \frac{\partial^2 u}{\partial x^2} + \dots$$

$$0 = a+b+c \quad [u_j]$$

$$1 = (-a+c) \Delta x \quad \left[\frac{\partial u}{\partial x}|_j \right]$$

$$0 = (a+c) \frac{\Delta x^2}{2!} \quad \left[\frac{\partial^2 u}{\partial x^2}|_j \right]$$

$$\Rightarrow a = -c \quad \Leftrightarrow a+c=0$$

$$a+b+c = (a+c) + b = 0 \Rightarrow b=0$$

$$\text{Then } 1 = (2c) \Delta x \Rightarrow c = \frac{1}{2\Delta x}$$

$$a = -\frac{1}{2\Delta x}$$

$$\begin{aligned}\frac{\partial u}{\partial x}|_j &= -\frac{1}{2\Delta x} u_{j-1} + \frac{1}{2\Delta x} u_{j+1} \\ &= \frac{u_{j+1} - u_{j-1}}{2\Delta x}\end{aligned}$$

$$\begin{aligned}\text{lead error} & (-a+c) \frac{\Delta x^3}{3!} \frac{\partial^3 u}{\partial x^3} \\ &= \frac{2}{2\Delta x} \frac{\Delta x^3}{3!} \frac{\partial^3 u}{\partial x^3} \\ &= \frac{\Delta x^2}{3!} \frac{\partial^3 u}{\partial x^3} \\ &= O(\Delta x^2)\end{aligned}$$

By following the above stated procedure, we can derive FD formula for any derivatives

Ex: Derive FD formula for $\frac{\partial u}{\partial x}$ using
five-point stencil $\{x_{j-2}, x_{j-1}, x_j, x_{j+1}, x_{j+2}\}$

Solⁿ

$$\left. \frac{\partial u}{\partial x} \right|_j = a u_{j-2} + b u_{j-1} + c u_j + d u_{j+1} + e u_{j+2}$$

$$= a \left[u_j - 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2!} \frac{\partial^2 u}{\partial x^2} - \dots \right]$$

$$+ b \left[u_j - \Delta x \frac{\partial u}{\partial x} + \dots \right]$$

$$+ c u_j$$

$$+ d \left[u_j + \Delta x \frac{\partial u}{\partial x} + \dots \right]$$

$$+ e \left[u_j + 2\Delta x \frac{\partial u}{\partial x} + \dots \right]$$

Now

$$a + b + c + d + e = 0 \quad (1)$$

$$(-2a - b + d + 2e)\Delta x = 1 \quad (2)$$

$$\left(\frac{\partial^2 u}{\partial x^2} \right): (4a + b + d + 4e) \frac{\Delta x^2}{2!} = 0 \quad (3)$$

$$\left(\frac{\partial^3 u}{\partial x^3} \right): (-8a - b + d + 8e) \frac{\Delta x^3}{3!} = 0 \quad (4)$$

$$\left(\frac{\partial^4 u}{\partial x^4} \right): (16a + b + d + 16e) \frac{\Delta x^4}{4!} = 0 \quad (5)$$

$$(3) + (4) \Rightarrow -4a + 2d + 12e = 0$$

$$-2a + d + 6e = 0 \quad (6)$$

$$(3) - (4) \Rightarrow 12a + 2b - 4e = 0$$

$$6a + b - 2e = 0 \quad (7)$$

$$b = 2e - 6a$$

$$\textcircled{5} \Rightarrow 16a + 2e - 6a + 2a - 6e + 16e = 0$$

$$12a + 12e = 0 \Rightarrow \boxed{a = -e}$$

then $b = -2a - 6e = -8a$

$$d = 2a + 6a = \cancel{8}a$$

Finally $-2a + 8a + 8a - 2a = \frac{1}{\Delta x}$

$$12a = \frac{1}{\Delta x}$$

$$a = \frac{1}{12\Delta x}$$

Hence $\frac{\partial u}{\partial x} \Big|_j = \frac{u'_{j-2}}{12\Delta x} - \frac{8u'_{j-1}}{12\Delta x} + 0 + \frac{8u'_{j+1}}{12\Delta x} - \frac{u'_{j+2}}{12\Delta x}$

$$= \frac{u'_{j-2} - 8u'_{j-1} + 8u'_{j+1} - u'_{j+2}}{12\Delta x}$$

Remark: It would be simpler if using skew-symmetry

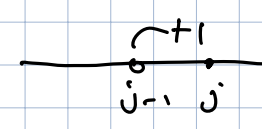
Second derivative

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial u'}{\partial x} \quad \text{if } u' = \frac{\partial u}{\partial x}$$

$$\approx \begin{cases} \frac{u'_{j+1} - u'_j}{\Delta x} & \text{if forward} \\ \frac{u'_j - u'_{j-1}}{\Delta x} & \text{if backward} \\ \frac{u'_{j+1} - u'_{j-1}}{2\Delta x} & \text{if central} \end{cases}$$

Replace u' to get u'' .

Centered approximation:

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \frac{u'_j - u'_{j-1}}{\Delta x} && \text{backward} \\ &= \frac{\frac{u_{j+1} - u_j}{\Delta x} - \frac{u_j - u_{j-1}}{\Delta x}}{\Delta x} \\ &= \frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2} && \text{(compact centered)} \end{aligned}$$


backward (forward) $\Rightarrow$ centered
forward (backward) $\Rightarrow$ centered

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \frac{u'_{j+1} - u'_{j-1}}{2\Delta x} \\ &= \frac{\frac{u_{j+2} - u_{j+1}}{2\Delta x} - \frac{u_{j+1} - u_{j-1}}{2\Delta x}}{2\Delta x} \\ &= \frac{u_{j+2} - 2u_{j+1} + u_{j-1}}{4\Delta x^2} && \text{(non-compact centered)} \end{aligned}$$

Re derive $\frac{\partial^2 u}{\partial x^2} \Big|_j$ using Taylor expansion.

$$u_{j+1} = u_j + \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2!} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^3}{3!} \frac{\partial^3 u}{\partial x^3} + \dots$$

(+)

$$u_{j-1} = u_j - \Delta x \frac{\partial u}{\partial x} + \dots$$

$$u_{j+1} + u_{j-1} = 2u_j + \frac{2\Delta x^2}{2!} \frac{\partial^2 u}{\partial x^2} + \frac{2\Delta x^4}{4!} \frac{\partial^4 u}{\partial x^4} + \dots$$

$$\frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2} = \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^2}{12} \frac{\partial^4 u}{\partial x^4}$$

Generalization

$$\frac{\partial^2 u}{\partial x^2} \Big|_j = a u_{j-1} + b u_j + c u_{j+1}$$

Collect coefficients. to get

H.W.

$$a + b + c = 0$$

...

Similarly

$$\frac{\partial^2 u}{\partial x^2} \Big|_j = a u_{j-2} + b u_{j-1} + c u_j + d u_{j+1} + e u_{j+2}$$

After using Taylor Series

$$a + b + c + d + e = 0$$

...

The calculation follows the steps as shown for $\frac{\partial u}{\partial x}$

Ex: Derive FD approximation of $\frac{\partial^2 u}{\partial x^2}$

Solⁿ

$$\frac{\partial^2 u}{\partial x^2} \Big|_j = a u_{j-1} + b u_j + c u_{j+1}$$

using Taylor series

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} \Big|_j &= a \left[u_j - \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2!} \frac{\partial^2 u}{\partial x^2} - \dots \right] \\ &\quad + b u_j \\ &\quad + c \left[u_j + \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2!} \frac{\partial^2 u}{\partial x^2} + \dots \right] \end{aligned}$$

Coeff of u_j :

$$0 = a + b + c$$

Coeff of $\frac{\partial u}{\partial x}$:

$$0 = (-a + c) \Delta x \Rightarrow a = c$$

Coeff of $\frac{\partial^2 u}{\partial x^2}$:

$$1 = (a + c) \frac{\Delta x^2}{2!} \Rightarrow a = \frac{1}{\Delta x^2}$$

$$0 = a + b + c \Rightarrow b = -(a + c) = -\frac{2}{\Delta x^2}$$

$$\begin{aligned} \text{Hence } \frac{\partial^2 u}{\partial x^2} &= \frac{1}{\Delta x^2} u_{j-1} - \frac{2}{\Delta x^2} u_j + \frac{1}{\Delta x^2} u_{j+1} \\ &= \frac{u_{j-1} - 2u_j + u_{j+1}}{\Delta x^2} \end{aligned}$$

Dispersion

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

Ex: Find dispersion relation of Centered FD Scheme for heat eqⁿ.

Solⁿ $u(x, t) = e^{i(kx + \omega t)} \quad \text{at } x_j = j \Delta x$

Scheme $\frac{\partial u_j}{\partial t} = \frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2}$

$x_{j+1} = x_j + \Delta x$

$$= \frac{e^{i(kx_{j+1} + \omega t)} - 2e^{i(kx_j + \omega t)} + e^{i(kx_{j-1} + \omega t)}}{\Delta x^2}$$
$$= \frac{e^{ik\Delta x} - 2 + e^{-ik\Delta x}}{\Delta x^2} e^{i(kx_j + \omega t)}$$

$$\frac{\partial u_j}{\partial t} = \frac{2\cos(k\Delta x) - 2}{\Delta x^2} u_j$$

$\frac{\partial}{\partial t} e^{i(kx_j + \omega t)}$ $i\omega e^{i(kx_j + \omega t)}$ $i\omega u_j$

$$= \frac{2(\cos \frac{2k\Delta x}{2} - 1)}{\Delta x^2} u_j$$
$$= \frac{2(-2\sin^2 \frac{k\Delta x}{2})}{\Delta x^2} u_j$$

$$i\omega u_j = -\frac{4}{\Delta x^2} \sin^2 \frac{k\Delta x}{2} u_j$$

$$i\omega = -\frac{4}{\Delta x^2} \sin^2 \frac{k\Delta x}{2}$$

Compare with exact

$$u(x,t) = e^{i(kx + \omega t)}$$

$$\frac{\partial^2 u}{\partial x^2} = -k^2 u(x,t)$$

$$\frac{\partial u}{\partial t} = i\omega e^{i(kx + \omega t)}$$
$$= i\omega$$

$$i\omega = -k^2$$

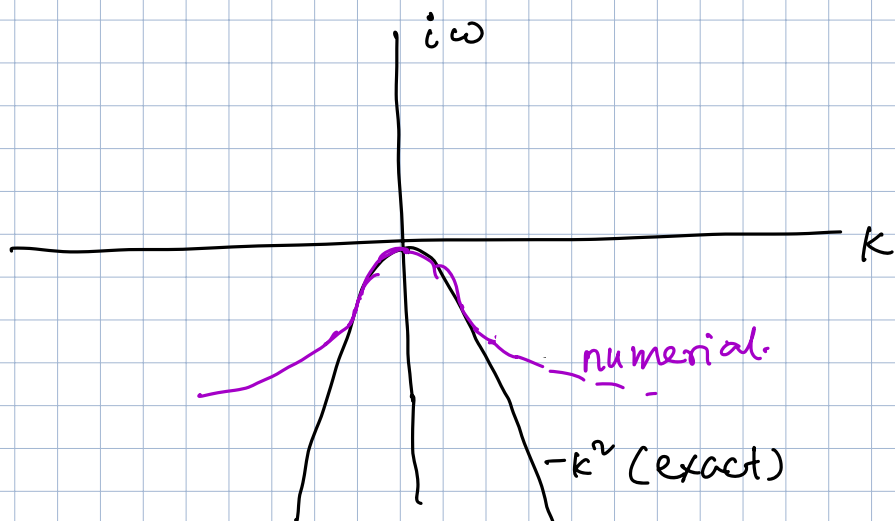
Using Taylor exp. of $\sin^2 \frac{k\Delta x}{2}$ we see that

$$-i\omega = k^2 - \frac{k^4 \Delta x^2}{12} + \dots$$

This suggest that $i\omega = -k^2$ for small Δx

Remark numerical scheme's dispersion approaches
to exact dispersion as $\Delta x \rightarrow 0$

Following plot compares the two cases.



Number of grid points

Dispersion relation of the scheme is

$\frac{2\pi}{\Delta x}$ periodic in k

Thus it is natural to take $k \in [-\frac{\pi}{\Delta x}, \frac{\pi}{\Delta x}]$

The number of grid points per wave length
is $\frac{2\pi}{k \Delta x}$.

